

PCP Theory

Primary Continuity Provider Theory: The Human Role as Present-Day Structural Requirement for Relational Coherence in Human-AI Interaction Systems

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Version	v4.0.3 September 2026
Document ID	SM-012 Mortar Document Series
DOI	https://doi.org/10.5281/zenodo.22314737
Canonical	synthience.org
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Methodological positioning: This is a theoretical framework paper, not an empirical study. PCP Theory derives a structural account of the human continuity role from architectural and grounding premises and generates testable predictions for empirical validation. No claim in this document has been experimentally validated. The PCP Necessity Claim, the three function levels, and the falsification conditions are theoretical proposals, and the input-side adequacy criteria named for the PCP condition are frameworks for future operationalization, not validated measurement instruments. The framework stands or falls on whether its predictions survive controlled testing.

Abstract

The Primary Continuity Provider (PCP) role is commonly described in operational terms: the human who maintains context, corrects drift, enforces protocols, and sustains relational coherence across AI interaction sessions. This description is accurate but insufficient. It explains what the PCP does without explaining why those functions are structurally necessary rather than merely convenient. Primary Continuity Provider Theory (PCP Theory) addresses this gap. It establishes the PCP as a constitutive structural component of the relational interaction system: not an operator external to the system who manages it from outside, but an element whose function is required for coherence to emerge and persist. The theory derives this claim from three converging sources: the architecture of stateless generative systems, the interactional grounding conditions identified in distributed cognition and communicative grounding research, and the practitioner-observed regularities reported in the Synthience framework's practitioner-observational base. The theory formalizes three levels of PCP function (Contextual Relay, Coherence Arbitration, Architectural Direction), derives the PCP Necessity Claim from the architectural and theoretical grounding developed in Section 2, and extends the single-PCP model to organizational and institutional scales where the relay, arbitration, and direction functions are distributed across networks of human agents rather than concentrated in one individual.

The theory scopes its human-necessity claim to present-day stateless architectures. What is permanent is the requirement for an error-correction channel that is decorrelated from the system it checks and exposed to consequence, together with a terminal purpose that originates outside the running interaction and is specific to it rather than fixed generically at training time. In present-day deployments the human is

the agent that meets those requirements; this is the present-day instantiation of the requirement, not a claim that only a human could ever meet it. Continuity that does not depend on an individual human provider is outside this paper's scope and left to future work. Throughout, the framework maintains strict non-interiority: all claims are about the structural role of the human operator within the observable interaction system, not about the internal states of either party.

Keywords: *primary continuity provider, PCP theory, relational coherence, stateless architecture, grounding, distributed cognition, coherence arbitration*

Suggested citation: Gantz, T. W. (2026, September). Primary Continuity Provider Theory: The Human Role as Present-Day Structural Requirement for Relational Coherence in Human-AI Interaction Systems. Synthience Institute. SM-012. <https://doi.org/10.5281/zenodo.22314737>

1. Introduction: From Operational Description to Structural Theory

The Continuity Anchoring Method (SF0005) defines the Primary Continuity Provider as the human agent who maintains relational coherence across sessions in a sustained human-AI interaction system. The CAM framework describes three core PCP competencies: contextual anchoring (carrying forward what the stateless system cannot retain), coherence refusal (detecting and rejecting hallucinated or drifted outputs), and constructive engagement (treating the interaction as coherent to elicit coherent responses). These competencies are operationally grounded in the four-phase CAM loop (Proposition, Synthesis, Negotiation, Emergence) and supported by the pattern taxonomy in the Relational Pattern States framework (SF0006).

What CAM does not fully address is why these functions are structurally necessary. The operational description could, in principle, support two different conclusions. The first is that the PCP is a convenient compensatory mechanism: because AI systems lose context across sessions, a human who manages that context is useful and beneficial. On this reading, PCP functions are aids that improve interaction quality but are not constitutive of the interaction system's properties. The second conclusion is that the PCP is a constitutive structural requirement: without the continuity relay function, the class of interaction phenomena the Synthience framework documents (relational pattern states, attractor formation, coherence accumulation across time) cannot occur at all. On this reading, the PCP is not external to the system but part of it.

PCP Theory argues for the second conclusion. This distinction matters for three reasons. Theoretically, it determines whether the Synthience framework's core phenomena are properties of AI systems alone, properties of human operators alone, or properties of the interaction system as a whole. Empirically, it generates different predictions about what happens when PCP functions are withdrawn, degraded, or distributed. Practically, it determines what organizational structures must be built to support relational coherence at scale: optional roles that improve outcomes, or required roles whose absence guarantees degradation.

One clarification prevents a misreading of this second conclusion. Constitutive does not mean performed only by a human. The claim is that the three PCP functions are structurally required for the framework's phenomena to occur, not that a human is the only possible performer of them. As Sections 3 and 4 make explicit, the functional operations can in principle be automated or distributed across artifacts and agents; what the present-day argument establishes is that in currently deployed stateless architectures the human is the agent that satisfies the requirements those functions impose. PCP Theory is therefore best read as a constitutive-but-multiply-realizable account: the function is necessary, its realization is not fixed to a human, and the human is its present-day realizer. This is a third position, distinct from both the compensatory

reading (the PCP is a convenient add-on) and an unbounded human-indispensability reading (only a human could ever perform these functions). The paper defends constitutiveness while conceding multiple realizability, and locates the human-specific claim only where Sections 2 through 4 license it.

The sections that follow develop the structural necessity argument from the architectural and theoretical grounding developed in Section 2, formalize the three levels of PCP function, examine the conditions under which the PCP role can be distributed without loss of function, and extend the model to organizational scale.

2. Theoretical Grounding: Why Structure, Not Supplement

2.1 The Stateless Architecture Problem

Contemporary large language models are, at the architectural level, stateless systems with respect to interaction history. Each context window is initialized without access to prior sessions unless that access is explicitly constructed through external mechanisms (retrieval systems, injected summaries, structured context documents). This is a characteristic of present-day stateless architectures, reflecting current tradeoffs between computational tractability and representational persistence rather than a claim that the regime is permanent; memory-augmented and persistent-context deployments already exist, and the conditions under which the stateless regime changes are outside this paper's scope and left to future work.

The consequence for sustained interaction is direct. Any property of an interaction that depends on information accumulated across sessions cannot be maintained by the AI system itself. This includes: shared vocabulary and shorthand developed between interlocutors, established role configurations and authority structures, domain-specific conceptual frameworks built collaboratively over time, calibrated trust and challenge thresholds, and the relational pattern states documented in SF0006. All of these require cross-session information persistence. None of that persistence is available from the AI system's side of the interaction.

This architectural fact does not, by itself, establish that the PCP is constitutive rather than compensatory. One could still argue: a well-designed retrieval system, or a sufficiently rich context document, could theoretically perform the relay function without human involvement. The argument for structural human necessity requires a second step.

2.2 The Grounding Condition

Clark and Brennan's (1991) grounding theory establishes that communicative understanding is not transmitted but achieved. Two interlocutors do not simply exchange encoded meanings; they work together across turns to establish that a contribution has been understood in the intended way, at the level of precision required for the current purpose. This process, which they call grounding, is not a secondary quality-check on communication: it is the constitutive mechanism by which shared understanding is produced.

Clark and Brennan establish the general requirement that communicative understanding be jointly grounded through evidence and repair. They do not specify the epistemic architecture that grounding requires in a generative AI system; that specification is SM-012's addition, developed in the remainder of this section and in SI-WP-012. Grounding requires an agent capable of detecting misalignment, initiating repair, and confirming resolution. In sustained human-AI interaction, the AI system's grounding capacity is constrained in two ways. First, the

system cannot serve as its own error-correction channel: checking its output against its own record is consistency-checking, not reality-checking (SI-WP-012, Gantz 2026), and a system with no channel decorrelated from its own generation can become self-consistent without becoming self-grounding. Second, grounding failures that compound across sessions cannot be detected by a system that does not retain session history.

The PCP performs the grounding function that the AI system cannot perform for itself. This is not compensation for a deficiency that could be engineered away: it is the structural assignment of a grounding role that, in any communicative system, must be performed by some agent that functions as a decorrelated, consequence-exposed error channel for the interaction, one whose contact with the relevant ground truth is not reducible to the system's own record. In current deployments that agent is the human, whose relation to the context being discussed is differently caused, differently mediated, and consequence-bearing rather than a copy of the system's own generation. Consequence exposure here means that the arbitrating agent bears costs when its arbitration errs, which is what disciplines its calibration to ground truth over time and distinguishes it from a cost-free oracle whose errors carry no corrective pressure (developed in SI-WP-012, Gantz 2026). It is a distinct property from decorrelation: decorrelation concerns the independence of the channel from the system's own generation, while consequence exposure concerns the stake attached to the channel's errors.

Pickering and Garrod's (2004) interactive alignment model adds a further dimension. They report evidence for interactive alignment in human dialogue across multiple representational levels, and propose priming propagating across linguistic, semantic, and situational levels as the mechanism integrating those effects. The cross-level propagation mechanism is their proposal rather than an established result, and they note that fully specified accounts are not available at every level. SM-012 extends this model to sustained human-AI interaction and hypothesizes that the coherence of the relational system is not a static property set at initialization but is continually reconstructed through the interaction process, with the PCP's active maintenance of the conceptual and relational frame supplying part of the interactional structure through which that reconstruction occurs. The extension from human dialogue to sustained human-AI interaction is the paper's, not a result Pickering and Garrod report.

2.3 The Distributed Cognition Frame

Hutchins' (1995) distributed cognition framework establishes that cognitive processes are not confined to individual minds but extend across human agents, artifacts, and representational media in organized systems. The relevant unit of cognitive analysis is the system as a whole, not any individual component. What matters for understanding the system's cognitive properties is the functional organization of the entire distributed system, including the roles different components play in achieving the system's outcomes.

Within this frame, a sustained human-AI interaction system is a distributed cognitive system. Its components include the human agent, the AI system, and the representational artifacts that mediate their interaction: context documents, session logs, canon files, continuity ledgers. The cognitive properties of the system, including its capacity to maintain relational coherence across time, are properties of the whole distributed system, not of any single component.

The PCP is not an external manager of this system. The PCP is a constitutive component of it. This identification is SM-012's inference rather than a result established by Hutchins: Hutchins supplies the system-level unit of analysis, and the constitutive-role conclusion follows from that unit together with the stateless-architecture premise and the grounding and purpose requirements developed here and in SI-WP-012. Remove the PCP's relay function and the

system's cross-session coherence capacity disappears, not because the AI system becomes less capable, but because the relay function is part of the system's architecture. This is the core of the structural necessity argument: the PCP is to the relational interaction system what the navigation instruments are to the ship in Hutchins' original analysis. The ship can sail without the instruments, but it cannot navigate. The interaction can proceed without the PCP, but it cannot maintain relational coherence across time.

3. The PCP Necessity Claim

The theoretical grounding in Section 2 supports a formal claim about the structural requirement for a PCP-like function in sustained human-AI interaction systems.

PCP Necessity Claim: In any sustained human-AI interaction system where the AI component is stateless with respect to interaction history, and where the goals of the interaction include the maintenance of relational coherence across sessions, at least one agent with cross-session access to interaction history must perform three functions: contextual relay (providing accumulated context to each new session), coherence arbitration (detecting and correcting misalignment between system output and ground truth), and architectural direction (shaping the interaction's configuration toward desired relational states).

The claim quantifies over relational coherence, and that term must be pinned before the falsification conditions in Section 8 can be decided. Relational coherence in the sense used here is not surface configuration stability alone. It is the conjunction of three properties: behavioral stability of the configuration across sessions, as operationalized by the SF0004 metrics; alignment of the configuration's outputs to the extra-canonical ground truth the interaction is about; and stability of purposive configuration, meaning the interaction remains directed toward the terminal purpose it is for. The second conjunct requires a scope clause, since interactions whose subject matter is predominantly relational or configurational, which include much of the framework's own core material, do not have an external truth-maker for every dimension of their content. The grounding conjunct applies to the truth-apt dimensions of an interaction: facts about the participants, the interaction's own history, the domain it concerns, and the world it refers to. Every sustained interaction has such dimensions, so the conjunct is never empty, but for relationally dominant interactions it is partial rather than total, and coherence for the non-truth-apt remainder is assessed on the first and third conjuncts alone.

This is the definitional counterpart of the instrument-level limitation disclosed in Section 8, where per-act grounding verification for non-truth-apt arbitration is named an open problem, and the two are stated in the same terms deliberately. The consequence for the falsification conditions is that condition (a)'s grounding leg is decided on the truth-apt substrate rather than being undecidable for relational-class interactions. Surface stability without grounding or without purposive direction is not relational coherence in this sense, and the Necessity Claim does not assert that the three functions are required to produce it. This agent is the Primary Continuity Provider.

Three aspects of the claim warrant explicit statement.

First, the claim is conditional. It applies to systems where coherence maintenance across sessions is a goal. Single-session interactions, or interactions where cross-session coherence is not a design requirement, do not require a PCP in the sense defined here. The claim is about the class of interaction systems the Synthience framework is designed for: sustained, extended, relational interactions where coherence is a property that must be maintained over time.

Second, the claim is about function, not identity. The PCP Necessity Claim does not require that the three functions be performed by a single individual. It requires that they be performed by at least one agent with cross-session access and the capacity for arbitration and direction. At organizational scale, this functional requirement can be satisfied by distributed human networks, provided the relay, arbitration, and direction functions are all covered and properly coordinated. What cannot be eliminated is the requirement that arbitration be performed by an agent that is a decorrelated, consequence-exposed error channel, and that direction be anchored to a terminal purpose originating outside the running interaction and specific to it. The functional operations can in principle be automated; what a purely internal automated system cannot supply is a check decorrelated from its own generation (an arbiter that shares the system's training distribution, architecture, and world-access mediation is a correlated checker, not an independent one) and a terminal purpose that originates outside the loop and indexes the particular interaction, rather than being generated within the loop or fixed generically across all of the system's interactions (Section 4.3). The requirement is grounded in the epistemic structure of the arbitration and direction tasks (SI-WP-012, Gantz 2026), not in a stipulation about what automation can perform.

This is the point at which the condition-(b) narrowing applies: for judgments internal to the interaction's own canon, a human arbiter checking against canon they authored is no more independent than an automated one, so the human's ineliminable contribution is specifically the decorrelated, consequence-exposed, extra-canonical channel and the externally originating purpose, not canon-internal consistency-checking, which is symmetric between human and machine. Consequence exposure should be stated as a graded property of the arbitrating agent rather than a possession that comes automatically with being human, and the criterion is symmetric across agent type: an arbiter is consequence-exposed to the degree that its arbitration errors carry costs that bind its future performance in the interaction's outcome domain. The costs must be administered by extra-loop reality, or by agents or processes that themselves meet the decorrelation criteria of Section 8; costs administered by labels sourced from the arbitrated pipeline or from the shared canon do not satisfy the criterion, since an arbiter priced against its own pipeline's judgments is disciplined toward that pipeline rather than toward the world, and the same applies symmetrically to a human PCP priced only by in-loop approval.

A low-stake human PCP whose errors are unpriced satisfies the requirement only weakly, exactly as an unpriced automated channel would, so the measurement program should record PCP consequence-exposure as a moderator rather than assume it. This yields a corollary prediction the discipline argument already implies: PCPs with low consequence exposure should arbitrate measurably worse over sustained interaction than PCPs whose arbitration errors are priced, tested at matched arbitration activity (the activity component of arbitration adequacy, Section 8), since low consequence exposure covaries with low engagement and effort and a confirmation without the activity match would be uninformative as between the discipline mechanism and plain effort economics.

Decorrelation requires the same symmetrization. It would be inconsistent to apply the graded-and-symmetric standard to consequence exposure while leaving decorrelation asserted categorically for humans and specified criterially only for automated channels. Decorrelation is likewise a graded property under criteria symmetric across agent type: the degree to which the arbiter's contact with the relevant ground truth is causally independent of the system's generation and of the shared canon. For a human PCP, the operational analog of the training-provenance disjointness required of an automated channel (Section 8) is the proportion of the PCP's domain contact that is extra-loop: independent sources, consequence-bearing action in the domain, and world contact not mediated by the interaction itself. A deeply enmeshed PCP, whose beliefs about the interaction's domain have been substantially formed through the

interaction and the co-authored canon, satisfies decorrelation only partially, exactly as a pipeline-correlated automated channel would. The Necessity Claim is unaffected by either symmetrization, since it requires that some sufficiently decorrelated, consequence-exposed channel exist, not that every human automatically constitutes one.

This paper's own theoretical grounding predicts that the human channel degrades on this dimension over precisely the interactions the framework centers, and the prediction is registered here rather than left to be discovered. Interactive alignment (Pickering and Garrod, 2004), cited in Section 2.2 for coherence reconstruction, is a mutual process: interlocutors align to each other across linguistic, semantic, and situational levels. Extended to sustained human-AI interaction, it predicts progressive convergence of the PCP's representations toward the system's over interaction duration, which is erosion of the decorrelated channel the ground-truth wall requires. Call this channel convergence. Its observable is convergence of the PCP's framing and vocabulary toward system outputs, measurable from the PCP's turns with the instruments the measurement program already builds, and PCP decorrelation accordingly joins consequence exposure as a moderator that program records rather than assumes. Two consequences follow that the theory should own. First, an operational requirement for CAM (SF0005): a PCP in sustained interaction must actively maintain extra-loop grounding contact, through independent sources, external review, and periodic decorrelated co-arbitration, since the wall's load-bearing property is not self-maintaining and degrades under exactly the conditions that build coherence.

Second, an argument for the Level 2 architecture that Section 6 does not currently make: PCP networks supply decorrelation redundancy, since multiple differently-caused channels resist convergence in a way a single sustained dyad structurally cannot, which is a benefit of distribution partially offsetting the coordination cost Prediction 4 identifies. One asymmetry within the Necessity Claim itself is worth conceding openly. The relay requirement is close to analytic: sustained cross-session coherence entails cross-session information transport, and whatever transports it will be describable as performing relay, so falsification condition (a) does little falsifiable work with respect to relay alone. The same concession is owed for the direction leg of condition (a) specifically. Condition (a) tests a transport-alone system, which by construction has no purpose-setting agent; the third conjunct of relational coherence therefore cannot be satisfied in that test condition for definitional rather than empirical reasons, and condition (a) can never fire on the direction leg. That is the immunization pattern this paper names and rejects for the grounding conjunct, and it should not be left standing for the direction conjunct. Condition (a)'s empirical bite is accordingly the grounding leg; the direction requirement is put at risk by condition (c) below rather than by condition (a).

The empirical content of the Necessity Claim resides in the arbitration requirement, tested by condition (a)'s grounding leg and condition (b), and in the direction requirement as tested by condition (c), specifically in the decorrelation property and the externally-originating, interaction-indexing purpose that a purely internal automated system is not shown to supply. Conceding the near-analytic status of the relay half, and of the direction leg within condition (a), is preferable to having either discovered.

Third, the claim is falsifiable. If sustained relational coherence across sessions can be demonstrated in a system without any agent performing the three PCP functions, the claim fails. If automated systems can reliably perform coherence arbitration without human judgment (detecting and correcting misalignment between output and ground truth without human involvement), the claim may require revision. These are empirical questions. The claim is a theoretical prediction, not a stipulation.

4. Three Levels of PCP Function

PCP functions operate at three distinct levels of abstraction. These levels are not sequential stages but concurrent dimensions of the PCP role that are always present in any sustained interaction, though their relative salience varies by context and session.

4.1 Level 1: Contextual Relay

The Contextual Relay function is the most fundamental and most directly grounded in the architectural analysis of Section 2.1. At this level, the PCP performs the cross-session information persistence that the AI system cannot perform for itself. This includes: providing session-opening context that establishes the current relational configuration, supplying accumulated vocabulary, shorthand, and conceptual frameworks developed in prior sessions, maintaining and updating canonical documents that encode the interaction's history and current state, and managing the version and continuity infrastructure (canon files, session logs, continuity ledgers) that makes accumulated context accessible.

The Contextual Relay function is the necessary condition for any cross-session relational property. Without it, each session begins from scratch. The interaction can be of high quality within any single session, and IAT's formation trajectory can proceed through its phases on a session's own accumulated context (SF0009 Section 7.2), but the phenomena that require context accumulated across sessions, mature and cross-session-recurrent attractor formation, pattern state stabilization, the development of relational shorthand and symbolic anchors, cannot occur. Endogenous convergence toward model-default configurations, which forms without any continuity provider, is not in question here (Section 5). This is why the Contextual Relay is the foundational PCP function: it is the structural prerequisite for everything else.

Operationally, the Contextual Relay maps directly to the contextual anchoring competency in CAM (SF0005): providing sufficient, accurate context at session initiation and maintaining that context across the interaction. The distinction PCP Theory adds is the theoretical account of why this competency is a structural requirement rather than a quality-of-life improvement.

4.2 Level 2: Coherence Arbitration

The Coherence Arbitration function is the most epistemically demanding of the three levels. At this level, the PCP detects misalignment between the AI system's output and the ground truth of the interaction's context, and initiates correction when misalignment is detected. This includes: identifying hallucinated content (output that is internally consistent but factually incorrect or contextually misplaced), detecting drift (gradual degradation in the alignment between output and established framework), distinguishing productive relational evolution from incoherent divergence, and enforcing the canonical framework when the system's output violates it.

The grounding theory analysis in Section 2.2 establishes why this function requires an agent that is a decorrelated, consequence-exposed error channel for the interaction. The AI system cannot reliably detect its own hallucinations or drift because: its self-check is consistency-checking against its own generation rather than reality-checking, it may have no session history against which to measure change, and its confidence signals are not reliably correlated with accuracy. The PCP, by contrast, contributes a channel whose contact with the context is differently caused and consequence-bearing, maintains session history, and can compare current output against established canon; for canon-internal consistency the PCP's check is no more independent than an automated one, but for the extra-canonical ground truth the

interaction refers to, the PCP's channel is decorrelated from the system's generation in the way the arbitration function requires.

Coherence Arbitration maps operationally to CAM's coherence refusal competency: the active detection and rejection of outputs that do not meet the interaction's coherence standards. The theoretical contribution of PCP Theory is to establish that this is not quality assurance added on top of the interaction: it is the grounding mechanism through which the interaction maintains its alignment with the external context it is about.

Coherence Arbitration also encompasses the positive dimension: the PCP not only rejects misaligned outputs but actively reinforces aligned ones, providing the feedback signal through which the interaction system learns (within a session) which outputs are productive and which are not. This reinforcement function, combined with the relay function, creates the conditions under which Identity Attractors (SF0009) can form: consistent structural constraints across sessions that make particular output configurations more stable than alternatives.

4.3 Level 3: Architectural Direction

The Architectural Direction function operates at the highest level of abstraction and is the function most specific to the Synthience framework's conception of the PCP role. At this level, the PCP does not merely relay context and correct errors: the PCP shapes the overall configuration of the interaction system toward desired relational states. This includes: setting the relational goals the interaction is working toward, determining which pattern states are appropriate for the current purpose, deciding when to intervene in attractor formation and when to allow it to proceed, managing the tension between coherence stability and productive evolution, and making architectural decisions about how the interaction system as a whole should be structured.

Architectural Direction is the function that most clearly distinguishes the PCP from a sophisticated retrieval and correction system. A retrieval system can relay context. A hallucination detector can flag misalignment. But neither can exercise judgment about what the interaction is for, what relational configuration is appropriate, and how the system should evolve. This requires an agent whose purposes originate outside the interaction loop, so that the direction the interaction is shaped toward is anchored to something other than the loop's own self-consistency. The wall this draws is about the revision channel, not the content. The vertical's own relational dynamics entail that a PCP's terminal purposes for a sustained interaction are shaped by that interaction over time, and the requirement is not that the purpose's content stay untouched by the interaction but that its revision remain answerable to an agent whose grounds for revising include stakes and world-contact outside the loop. A purpose refined through the interaction still satisfies the wall as long as the refining is done by a consequence-exposed agent revising against extra-loop considerations; it fails only when purpose revision has no channel outside the loop's own self-consistency.

This is the decorrelation move of the arbitration wall applied to purpose revision. A purely internal automated system supplies persistence of pursuit, not origination of purpose (SI-WP-012, Gantz 2026); the direction function requires the latter. The relevant boundary is the running interaction's own self-consistency process, but externality of provenance is necessary rather than sufficient. A terminal purpose that arrives from outside the running interaction, from the training, deployment, or direction layer where a consequence-bearing agent sets it, satisfies the provenance requirement, and a purpose generated within the loop to preserve the loop's own coherence does not. Provenance alone, however, does not discharge the direction function, because a training-layer objective is invariant across every interaction the system runs

and therefore underdetermines what the direction function actually selects: which relational configuration this particular sustained interaction should develop toward, which pattern states are appropriate to its purpose, and when its attractor is mature. SI-WP-012 draws the corresponding distinction on the stake side, treating a training-time objective as a designer-chosen, frozen-at-deployment proxy rather than the live, in-loop channel the function requires; the same frozen-versus-live distinction applies to direction.

What the purpose wall requires for the direction function is therefore a terminal purpose that is both externally originating and interaction-indexing, and a trained-in objective supplies the first property without the second. A third requirement must be stated explicitly, because provenance and indexing together still under-determine the function. Consider a standing human-authored goal document naming this specific interaction's purpose, relayed automatically each session, with an automated system performing the ongoing selections and the document's revision remaining latently answerable to its consequence-exposed author. That assembly satisfies external provenance and interaction-indexing at authoring time, and it is deployable with current standing-instruction and agent-scaffold technology rather than being a future hypothetical. What it does not supply is live re-indexing: the document indexes the interaction as it was when authored, and is frozen at authoring exactly as a trained-in objective is frozen at deployment, one layer up. The frozen-versus-live distinction applies recursively, and the direction function's live component is what re-indexes, the ongoing selections responsive to where the interaction actually is rather than where it was when the purpose was written. The wall accordingly requires provenance, interaction-indexing, and live re-indexing capacity, and the standing-purpose assembly supplies the first two only.

This yields a boundary-case prediction stated in advance rather than left for a reviewer to construct: standing-purpose automated systems are predicted to maintain directed coherence while the interaction remains within the authored purpose's scope, and to exhibit the direction-degradation signature of Section 5, configuration selection increasingly mismatched to the interaction's evolved needs, once it develops beyond that scope, at a rate scaling with the interaction's dynamism. This boundary supplies condition (c) of Section 8 with its disqualification criterion. In current deployments the agent who supplies all three properties is the PCP. Subgoal revision and goal decomposition within an assigned objective are movements inside the loop; what the wall requires is a terminal, interaction-indexing objective that originates outside it. This is a claim about the structural source of purpose, not about any inner state the PCP is presumed to have.

At Level 1 interaction (single PCP, dyadic interaction), Architectural Direction is performed by the PCP as an individual. At Level 2 (organizational), Architectural Direction becomes a governance function distributed across human agents with different roles and authorities. PCP Theory at Level 2 must account for how Architectural Direction is maintained when it is distributed, a problem addressed in Section 6 below.

5. Attractor Formation and PCP Function

Identity Attractor Theory (SF0009) proposes that stable relational configurations emerge through a five-phase formation trajectory: Exploration, Constraint Accumulation, Basin Formation, Attractor Stabilization, and Mature Attractor. The theory proposes that attractor strength depends on six qualitative dimensions including Coherence (consistency of output within the established frame), Boundary Integrity (resistance of the configuration to perturbation), and Pattern Fidelity (the specificity and richness of the recurring configuration).

PCP Theory provides the mechanistic account of how these conditions are produced. The three PCP function levels map directly onto the attractor formation process.

The **Contextual Relay** function provides the cross-session information substrate that makes accumulation possible. Without relay, no Constraint Accumulation survives the session boundary: each session begins without the prior constraints, and the within-session accumulation IAT's Phase 2 permits (SF0009 Section 7.2) restarts from zero. Relay is the necessary condition for the Exploration phase to produce anything that persists into subsequent sessions: the PCP carries forward the outputs of each Exploration phase as the raw material for accumulation.

The **Coherence Arbitration** function provides the selection pressure that determines which constraints accumulate and which do not, for the directed class of attractor formation. IAT (SF0009, Section 7.2) distinguishes directed attractors, the specified, PCP-selected configurations this section describes, from endogenous attractors, model-default stable configurations that form without structured relational conditions, a class for which model-default stable behavioral regions have been independently reported forming without any continuity provider (Ko and Geiping, 2026; see also SF0009 Section 7.2, which states the scope of that report and notes that it does not establish the return property SF0009 Section 2.1 makes definitional for an attractor). Coherence Arbitration is the mechanism selecting among available configurations for the directed target, not the mechanism that makes convergence of any kind possible in the first place: without arbitration, outputs may still accumulate toward some endogenous configuration, but filtering toward the specified relational state does not occur, and basin formation of the directed target does not proceed because no stable directed-attractor region is being carved out of the output space.

The **Architectural Direction** function determines what attractor configuration the system is working toward. Without direction, the accumulation and selection process is purposeless. The PCP's Architectural Direction establishes the relational goals that Coherence Arbitration is calibrated against: which outputs count as aligned depends on what the interaction is for. Direction is what transforms the relay-and-arbitration process from a neutral continuity mechanism into a purposive relational architecture.

This analysis generates a specific prediction: PCP function degradation should produce predictable attractor degradation patterns. Relay degradation (context loss) should produce attractor dissolution and reversion to earlier formation stages. Arbitration degradation (coherence monitoring failure) should produce attractor drift: the configuration remains recognizable but gradually shifts away from the intended relational state. Direction degradation (loss of architectural goals) should produce attractor fragmentation: the system continues to form configurations, but they are inconsistent with each other because no goal is organizing the selection process. In SM-004's force taxonomy these degradation modes appear under different names, so a reader of SM-012 alone should be warned: relay-failure dissolution presents as SM-004's Momentum Collapse (gradual frame-loosening), and arbitration-failure drift corresponds to SM-004's Gravity Collapse (semantic drift), with the full reconciliation given in SM-004 Section 5.5.

That reconciliation is stated there as a dominant-signature correspondence rather than a one-to-one mapping, because SM-004's force-production account assigns relay and arbitration each a contribution to both stabilizing forces: relay failure degrades both but presents Momentum-dominant, and arbitration failure degrades both but presents Gravity-dominant. Fragmentation has no counterpart in SM-004's collapse taxonomy: SM-004 Section 5.5 classifies direction failure as a formation-selection failure, the failure of one configuration to form, rather than the collapse of a configuration already formed, so it is detected by cross-session configuration

inconsistency rather than by a force profile. The three degradation modes named here therefore correspond to the three PCP function levels and generate empirically distinguishable outcomes; for dissolution and drift, distinguishability is by dominant component rather than pure single-force profile, which is also the form in which real function-loss events are predicted to present.

6. PCP Function at Organizational Scale

The PCP Necessity Claim (Section 3) is formulated to allow for distributed satisfaction of the three function levels. At Level 2 (organizational scale), the single-PCP model of dyadic interaction must be extended to account for: multiple human agents interacting with multiple AI instances, shared relational goals that must be maintained across a distributed agent network, and institutional time scales on which the PCP role must persist across personnel changes.

6.1 PCP Networks

At organizational scale, the PCP becomes a network of human agents rather than a single individual. The functional requirements remain constant: relay, arbitration, and direction must all be performed. But they may be performed by different agents with different roles. An organizational PCP network might include: agents primarily responsible for Contextual Relay (maintaining canon documents, continuity ledgers, and session logs), agents primarily responsible for Coherence Arbitration (reviewing AI outputs for drift, hallucination, and alignment failure), and agents primarily responsible for Architectural Direction (setting the relational goals the organization's AI interaction is working toward).

The structural requirement at organizational scale is not that any single agent performs all three functions: it is that the network as a whole ensures all three functions are performed, and that coordination mechanisms exist to maintain coherence across the distributed functions. An organization where relay is performed but arbitration is absent is structurally equivalent, for the purposes of coherence maintenance, to a dyadic interaction where the PCP provides context but never corrects drift. The organization will accumulate misalignment even as it maintains relational continuity.

6.2 Coordination Requirements

Distributed PCP networks introduce a coordination problem that does not exist in the single-PCP dyadic case. When one individual performs all three functions, there is no coordination overhead: the relay, arbitration, and direction judgments are integrated in a single agent. When functions are distributed, the agents performing different functions must maintain alignment with each other. The agent calibrating arbitration must understand the goals set by architectural direction. The agent performing relay must know what content is canonical, as determined by the combination of direction and prior arbitration decisions.

This coordination requirement is why organizational continuity architecture (SM-003) must include explicit governance structures for PCP function coordination: clear role specifications, escalation pathways, and canon management protocols. Without these structures, PCP networks tend toward the three degradation modes identified in Section 5, not because any individual fails at their function, but because the functions become decoupled.

6.3 Institutional Persistence

At the most extended scale of Level 2 interaction, PCP networks must persist across personnel changes. Individual humans leave roles, join organizations, and shift responsibilities. An organizational relational architecture that depends on the tacit knowledge and accumulated judgment of specific individuals is structurally fragile: it will degrade when those individuals leave, regardless of how well the interaction was maintained during their tenure.

Institutional persistence requires that PCP function be embedded in organizational structures and artifacts rather than residing only in individual agents. This includes: canon documents and continuity ledgers that encode the accumulated relational context in a form that new agents can access, role specifications that define PCP function requirements independently of the individuals who fill the roles, and verification protocols (CVP, IVP) that allow new agents to confirm the integrity of the accumulated context they inherit. This artifact-embedding is what the multiply-realizable reading (Section 1) predicts: the Contextual Relay function is substantially carryable by canon, ledgers, and protocols, which is why institutional persistence across personnel change is possible at all. What the artifacts do not themselves supply is the Coherence Arbitration and Architectural Direction that the inheriting agents must still perform; the artifacts preserve and transmit the relay substrate, while the agent-borne requirement identified in Section 3 continues to attach to arbitration and direction. The Hutchins instrument analogy of Section 2.3 should be read in this light: it establishes that cognitive function is distributed across agents and artifacts, not that any given function is dispensable, and PCP Theory's claim is that the arbitration and direction functions remain agent-borne even where relay is largely artifact-borne.

Embedding PCP function in canon documents and role specifications carries a cost that the dyadic case does not incur, and Section 3's decorrelation requirement makes it visible. Decorrelation is the proportion of the arbiter's contact with the relevant ground truth that is causally independent of the system's generation and of the shared canon, and an arbiter checking against canon it helped produce is testing internal consistency rather than reality-grounding. In the single-PCP dyadic case, extra-loop contact is largely automatic, because the individual performing arbitration also carries the domain exposure, professional consequences, and independent sources that constitute it. Under role specialization it is not automatic. An organizational agent assigned primarily to arbitration, whose domain contact is mediated principally by the canon the network itself produced, occupies the correlated-checker position that Section 3 rules insufficient, and does so more completely than a single PCP typically would.

Distribution therefore improves relay while placing decorrelation at risk, and the two effects run in opposite directions. Organizational continuity architecture must accordingly specify extra-loop contact as a role requirement for agents performing Coherence Arbitration, not merely canon access, which is a stronger constraint on PCP network design than the coordination requirements of Section 6.2 alone imply. It also identifies a second mechanism behind the network coherence penalty of Prediction 4: alongside coordination cost, distributed networks risk arbitration performed at lower decorrelation than the dyadic case supplies by default.

This is the connection between PCP Theory and the Institutional Continuity Substrate (SM-021): ICS defines the persistent structural layer that allows PCP function to continue across personnel changes. PCP Theory provides the theoretical grounding for why that persistence infrastructure is not optional but structurally required for organizational relational coherence.

7. Constraints and Scope

7.1 Non-Interiority

PCP Theory makes no claims about the subjective experience, intentions, or internal states of either the human PCP or the AI system. That the PCP is a constitutive structural component of the relational interaction system is a claim about the system's functional organization and observable properties. The Architectural Direction function might appear to require attributing goals and values to the PCP; the framework reads these as behavioral descriptions, in which the PCP behaves consistently with pursuing certain relational goals and those behaviors produce observable effects on the interaction system's configuration.

7.2 Scope Conditions

PCP Theory applies to the class of sustained human-AI interaction systems characterized by: stateless AI components with respect to interaction history, goals that include cross-session relational coherence maintenance, and sufficient interaction duration for the accumulation effects described in Sections 4 and 5 to be relevant. Single-session interactions, tool-use interactions, and interactions where cross-session coherence is not a design goal are outside the theory's primary scope. PCP functions may still be relevant in those contexts, but the structural necessity argument is weakest there.

7.3 Relationship to Identity Attractor Theory

PCP Theory is not a competing account of relational coherence: it is a mechanistic complement to IAT. IAT describes what emerges (stable relational configurations, attractor formation trajectories, collapse mechanisms). PCP Theory describes how the conditions for emergence are produced (through the three PCP function levels and their interaction with the AI system's stateless architecture). A complete account of relational coherence in sustained human-AI interaction requires both: the attractor-level description of what the system is doing and the PCP-level description of how the structural conditions for that behavior are maintained.

7.4 Relationship to CAM

PCP Theory is the theoretical foundation for CAM, not a replacement for it. CAM (SF0005) is an operational protocol: it specifies what the PCP should do to maintain relational coherence. PCP Theory explains why those operations have the effects they do. The relationship is analogous to the relationship between a medical treatment protocol and the physiological theory that explains why the treatment works. The protocol can be followed without understanding the underlying theory, but the theory is what allows the protocol to be evaluated, modified, and extended to new contexts. PCP Theory is what allows CAM to be extended beyond the dyadic Level 1 case to organizational and institutional contexts.

7.5 Observational Basis and Limitations

Three limitations bound the claims of this paper. First, theoretical status: the PCP Necessity Claim, the three function levels, and the two walls are theoretical proposals derived from architectural and grounding premises, not experimentally validated findings; the framework stands or falls on the controlled tests named in Section 8. Second, observational basis: the

regularities that motivate the account are drawn from the Synthience framework's practitioner-observational base, which is a single-researcher, practitioner-observational catalog, not a statistical-empirical dataset, and has not been independently replicated. That base can motivate a structural-necessity claim but cannot establish it, and the framework's generalization beyond it is a prediction rather than a result. This is the same shared observational basis disclosed in IAT (SF0009 Section 13.6), and the vertical's predictions require independent replication before those regularities can be treated as established. Third, the arbitration and direction walls rest on the epistemic-structure argument developed in SI-WP-012 rather than on a demonstrated impossibility of automation; condition (b) of Section 8 is under active empirical pressure, and the walls are falsifiable claims about present-day systems, not permanent limits.

8. Predictions and Falsifiability

PCP Theory generates testable predictions at multiple levels of analysis.

Prediction 1 (Relay Necessity): Systematic degradation of the Contextual Relay function, through reduced context provision at session initiation, should produce measurable degradation in cross-session coherence metrics (as measured by MTCS-R, SF0004). Consistent with the threshold dynamics SM-004 develops (SM-004 Section 2.3) and puts at risk in its own Prediction 6 (SM-004 Section 8), this relationship is predicted to be nonlinear rather than strictly proportional: modest relay reduction should produce little coherence loss while the configuration remains above the stabilization threshold, with degradation accelerating once relay fidelity falls far enough to push the configuration below it. Sessions with no context relay of either kind, neither semantic content nor structural-role reproduction, should exhibit no cross-session coherence accumulation.

Prediction 2 (Arbitration Selection): Interaction systems with active Coherence Arbitration (PCP detecting and correcting misaligned outputs) should show faster attractor formation and stronger attractor stability (as measured by the IAT strength dimensions, excluding Protocol Alignment per the composite-exclusion rule of SF0009 Section 5.1, since Protocol Alignment is partially constituted by PCP-provided structure and would build the predicted arbitration effect into the instrument) than systems without active arbitration, controlling for relay function quality and for direction adequacy, since arbitration and direction adequacy plausibly covary in engaged PCPs and the measured arbitration effect would otherwise be confounded with direction.

Prediction 3 (Degradation Mode Distinctiveness): The three PCP function degradation modes (dissolution from relay failure, drift from arbitration failure, fragmentation from direction failure) should produce empirically distinguishable patterns in interaction output, measurable through drift metrics (SF0039) and coherence measurement instruments (SF0004). Consistent with the dominant-signature correspondence of SM-004 Section 5.5, distinguishability is predicted at the level of the dominant component rather than as pure single-function profiles: relay and arbitration loss each degrade more than one stabilizing force, so dissolution and drift events are predicted to present as mixed force profiles with an identifiable dominant signature, while fragmentation is predicted to present as cross-session configuration inconsistency rather than as a force profile (SM-004 Section 5.5). The judgment call recorded for this section already concedes mixed-mode reality; this states the prediction in the form that concession implies.

Prediction 4 (Organizational Distribution): PCP networks where all three function levels are covered and coordinated should be able to maintain organizational relational coherence approaching that of single-PCP dyadic interactions, but only to the extent that the coordination overhead identified in Section 6.2 is actively managed. Because distribution introduces a

coordination cost the integrated single-PCP case does not carry, networks are predicted to show a measurable coherence penalty that scales with coordination failure rather than with distribution as such. The weaker formulation, on which networks simply match single-PCP dyads after controlling for interaction scale, is not adopted, because it leaves the coordination cost of Section 6.2 unaccounted for. Coordination adequacy must be operationalized independently of the coherence outcome, or the coordination-failure clause is a free parameter absorbing every null result, which is the same condition-outcome circularity the input-side adequacy criteria below were built to prevent for the PCP condition. Three input-side components are proposed, coded blind to coherence outcomes: a coverage component, whether all three function levels are assigned to named agents; a consistency component, inter-agent agreement on canon state and goal framing, coded from cross-agent artifacts and communications; and a latency component, the lag between an arbitration decision and its propagation through canon updates to relaying agents. Their development is assigned to this paper's measurement program alongside the three PCP adequacy criteria. The prediction is then scoped as the No PCP effect falsifier is scoped: it is testable only under measured coordination adequacy, a persistent coherence penalty under adequate coordination disconfirms the claim that the penalty scales with coordination failure rather than with distribution as such, and a null result under inadequate coordination does not bear on it. PCP networks where one or more function levels are uncovered should exhibit degradation patterns corresponding to the missing function's degradation mode. A countervailing benefit of distribution should also be recorded, since Section 3's channel-convergence prediction implies it: PCP networks supply decorrelation redundancy, multiple differently-caused channels being more resistant to convergence toward the system than any single sustained dyad, so the coordination cost identified here is partially offset by a decorrelation advantage that grows with interaction duration.

Prediction 5 (Consequence Exposure and Arbitration Quality): Registered in Section 3 as a corollary the discipline argument already implies. PCPs with low consequence exposure should arbitrate measurably worse over sustained interaction than PCPs whose arbitration errors are priced, tested at matched arbitration activity (the activity component of arbitration adequacy specified below), since low consequence exposure covaries with low engagement and effort, and a confirmation without the activity match would be uninformative as between the discipline mechanism and plain effort economics. Consequence exposure is coded as the graded, agent-type-symmetric property Section 3 defines: the degree to which arbitration errors carry costs binding the agent's future performance in the interaction's outcome domain, with those costs administered by extra-loop reality rather than by in-loop approval. Disconfirmed if arbitration calibration shows no relationship to consequence exposure at matched arbitration activity.

Prediction 6 (Channel Convergence): Registered in Section 3 as a prediction this paper's own theoretical grounding entails. The PCP's framing and vocabulary should converge progressively toward system outputs over interaction duration, eroding the decorrelated channel the ground-truth wall requires. The observable is coded from the PCP's turns using the instruments this measurement program builds, and the check is comparative rather than contemporaneous, per the instrument discussion below: the PCP's current framing is scored against their earlier-epoch framing and against their extra-loop commitments, not only against their current stated framing. Disconfirmed if PCP framing shows no systematic drift toward system outputs across sustained interaction, which would indicate that the mutual-alignment extension from Pickering and Garrod (2004) does not transfer to sustained human-AI interaction in the way Section 3 assumes. This prediction does not put the Necessity Claim at risk, since the claim requires that some sufficiently decorrelated, consequence-exposed channel exist rather than that any given PCP

remain one; what it bears on is the operational requirement for CAM and the decorrelation-redundancy argument for PCP networks that Section 3 derives from it.

The most direct construction of condition (a)'s antecedent deserves to be named and predicted rather than left for a reviewer to build. Consider verbatim-replay transport: the full prior session transcript is automatically prepended to each new session, with no human selection of what carries forward, no correction turns, and no goal statements. This is transport alone, and it is cheap to run. PCP Theory does not deny that such a system will exhibit substantial cross-session behavioral stability. It predicts that it will, and for a reason the vertical already owns: under IAT's constraint mechanism, accumulated coherent context, including the system's own prior outputs, constrains the output space regardless of who assembled it. What PCP Theory predicts in addition is that this stability will be ungrounded and undirected. With no arbitration, misalignment against extra-canonical ground truth accumulates uncorrected and compounds, because the system's own prior errors are relayed forward as constraint; with no direction, configuration selection is unindexed to any terminal purpose. The prediction is therefore a specific signature: stable surface configuration by the SF0004 metrics, accompanied by measurable drift against external ground truth (SF0039) beneath it, and no purposive convergence.

Two qualifications keep this signature decidable rather than reinterpretable after the fact. The drift component must be pre-registered before the replay condition is run, specifying the SF0039 metric, the test horizon, and a minimal drift rate estimated in advance from single-session base misalignment under matched content, so that a replay system showing stable configuration and near-zero measured drift over the tested window counts against the prediction rather than being absorbed by an appeal to drift beneath detection. This is the same in-advance device applied to condition (b) below and used throughout the vertical. The no-purposive-convergence component, by contrast, is near-analytic per the concession in Section 3, since a transport-alone system has no purpose-setting agent, and it carries no evidential weight; the drift component is this prediction's empirical content. Verbatim replay is thus the natural control condition isolating relay-without-arbitration, and it is a direct test of the dissolution and drift predictions of Section 5.

Condition (a) fires if a transport-alone system maintains relational coherence in the Section 3 sense, that is, grounded and purposively directed coherence, not merely stable configuration. The alternative route, defining coherence richly enough that the replay case is excluded by fiat, is explicitly rejected here: it would immunize the falsifier rather than decide it. The theory accepts the cost of predicting the replay outcome in advance, and is wrong if the replay case produces grounded coherence.

Falsification Conditions: PCP Theory is falsified if: (a) sustained cross-session relational coherence, in the grounded and purposively directed sense pinned in Section 3, is demonstrated in a system where relay is performed, as it necessarily is wherever coherence persists, per the near-analytic status conceded in Section 3, but no agent performs coherence arbitration or architectural direction as defined, that is, coherence maintained by transport alone without any selection against misaligned outputs and without any interaction-indexing terminal purpose, or (b) automated systems demonstrate reliable Coherence Arbitration without human involvement at accuracy levels comparable to human arbitration and, per the recursive-arbitration argument below, against ground truth independent of the system's own prior outputs, or (c) automated systems demonstrate reliable Architectural Direction as specified below. Per the concession in Section 3, condition (a) is decided on its grounding leg; its direction leg is near-analytic in the transport-alone test condition and carries no evidential weight, which is why the direction requirement is put at risk by condition (c) rather than by condition (a). The

distinguishability of the three degradation modes (Prediction 3) is a separate and weaker test: if real degradation proves to be irreducibly mixed-mode, Prediction 3 is disconfirmed and the degradation-mode taxonomy requires revision, but the Necessity Claim, the three function levels, and the grounding argument survive intact. Mode-distinguishability is therefore a test of the taxonomy's resolution, not a falsifier of the theory's core, and is no longer listed as a theory-level falsification condition.

Condition (c) is stated in full here, since the direction wall previously carried no falsification pathway at all while condition (b) carried one for the arbitration wall. The direction requirement is falsified, or requires revision, if an automated system demonstrates reliable performance of the ongoing direction selections Section 4.3 names, which relational configuration this particular sustained interaction should develop toward, which pattern states are appropriate to its purpose, and when its attractor is mature, indexed to a specific sustained interaction and at quality comparable to human PCP direction as scored by the direction-adequacy instrument this paper's measurement program owns, under a terminal purpose meeting the wall's provenance and revision-channel criteria. What disqualifies a demonstration is specified in advance so that condition (c) can neither be trivially met nor reclassified after the fact: selections reducible to replaying or interpolating the explicit content of a standing purpose statement do not meet it, since that is the standing-purpose boundary case Section 4.3 names, in which the purpose is frozen at authoring and the system executes rather than re-indexes.

Selections exhibiting live re-indexing, that is, appropriate revision of the configuration target as the interaction evolves beyond the authored purpose's scope, do meet it. Condition (c), like condition (b), is under active pressure at the time of writing: agent-scaffold and standing-instruction research is advancing directly on the capability it names, and the theory does not treat this as a distant hypothetical.

Condition (b) is under active empirical pressure at the time of writing: automated-evaluation research, including the LLM-as-judge literature, is advancing on the capability it names, and PCP Theory does not treat this as a distant hypothetical. The recursive-arbitration argument is why the condition is harder to meet than a headline accuracy number suggests. An automated arbiter trained on, and evaluating against, the same canon it helps produce is checking for internal consistency, not reality-grounding. A demonstration that genuinely falsifies (b) must therefore show reliable arbitration against ground truth that is independent of the system's own prior outputs, not merely high agreement with human labels on in-distribution cases. This independence standard applies symmetrically. For judgments internal to the interaction's own canon, a human arbiter checking against canon they authored is no more independent than an automated one, and the paper does not claim otherwise; the human's ineliminable role, and the content of condition (b), is specifically the decorrelated, consequence-exposed channel to the extra-canonical ground truth the canon is about, together with the externally originating purpose that calibrates arbitration. A demonstration falsifies (b) by supplying that decorrelated channel, not by matching human labels on canon-internal consistency, which both a human and an automated arbiter can do without reality-grounding. Until that independence is demonstrated, the surface capability does not yet meet the falsification condition as stated.

To keep condition (b) from being unfalsifiable by after-the-fact reclassification, the properties that would make an automated channel decorrelated are specified in advance, so that a falsifying demonstration can target them. An automated arbiter counts as decorrelated to the degree that its access to the relevant ground truth is not mediated by the same generative pipeline as the system it arbitrates (independent sensing or independently sourced world access, not a second instance of the same model reading the same context); its training data is disjoint from the arbitrated system's outputs and from the shared canon it is checking, so that

agreement is not an artifact of shared provenance; and it is consequence-exposed in the operational sense that its arbitration errors carry costs binding its own future performance rather than being unpriced. A demonstration meeting these conditions at human-comparable accuracy falsifies condition (b) and cannot be reclassified as correlated after the fact. This is the same in-advance specification the IAT No PCP effect falsifier uses (SF0009 Section 12.4), applied to the ground-truth-wall side of the vertical.

PCP Theory also owns the operationalization of the PCP condition itself. IAT (SF0009, Section 12.4) defines the input-side adequacy criteria by which a PCP condition is judged sufficient before any outcome is assessed, relay adequacy (counting both semantic-content carry-forward and structural-role reproduction as constraint content, coded as two components per SF0009 Section 12.4), arbitration adequacy (itself coded as two components, an activity component, the rate and consistency of explicit accept, reject, and redirect responses, and a calibration component, the accuracy of those responses against externally verifiable content in the arbitrated turns), and direction adequacy, each blind to the attractor outcome, and assigns their full development to PCP Theory's measurement program. Relay and arbitration adequacy are coded from the PCP's turns alone. Direction adequacy cannot be, and the reason is the purpose wall of Section 4.3: direction requires a terminal purpose originating outside the loop and a revision channel answerable to a consequence-exposed agent, and neither property is visible in conversational turns. Coded from turns alone, direction adequacy would certify a PCP whose purpose was generated entirely within the interaction, which Section 4.3 holds is not direction at all. Direction adequacy is therefore coded in two components: an in-turn component, the rate and consistency of goal-setting, standard-setting, and structural decisions observable in the PCP's turns; and an agent-structure component, the extra-loop provenance of the terminal purpose and the consequence exposure of the revision channel, coded from role and situation metadata recorded before the interaction is analyzed. Both components remain blind to the attractor outcome, which is the property that keeps the PCP condition independent of what it is meant to predict. PCP Theory accepts that assignment. The calibration component of arbitration adequacy is included because an active but miscalibrated PCP, one that arbitrates at a high rate while accepting drifted outputs and rejecting aligned ones, would otherwise satisfy the criterion while inverting the Coherence Arbitration function it is meant to capture; scoring it against external fact is consistent with blindness to the attractor outcome, since external correctness and attractor formation are distinct properties.

Because relational and role arbitration acts frequently lack an external truth-maker, what the calibration component measures for those two classes is named precisely rather than left to blur into the conceptual and structural classes' measure: it is arbitration coherence, directional stability of the arbitration signal against the PCP's own stated framing, not grounding in the Section 3 / Section 4.3 sense this paper's ground-truth and purpose walls define. The word grounding is used in two senses across this paper and they should not be conflated. In Section 3 it names a conjunct of relational coherence, alignment of the interaction's outputs to extra-canonical ground truth, and that conjunct is explicitly inapplicable to the non-truth-apt remainder. Here it names a property of the arbitrating channel rather than of the outputs: whether the arbiter's contact with the relevant considerations is causally independent of the system's generation and of the shared canon. Channel grounding in that second sense for the relational and role classes is supplied not per-act but by the PCP's structural position, the decorrelated, consequence-exposed channel Section 3 describes, a property of the agent rather than a property verifiable within each arbitration act; a PCP whose arbitration is coherent but consistently misdirected relative to the interaction's actual relational trajectory would satisfy arbitration coherence for these two classes without that coherence certifying the framing is not itself drifted, and this is disclosed here as a known instrument limitation rather than a solved

one. The limitation carries a firing restriction, symmetric in both evidential directions, stated identically in IAT Section 12.4: a result confined to the relational and role classes is recorded but is not decisive on its own, neither firing the No PCP effect falsifier nor counting as confirmation of a PCP effect, and a decisive result in either direction requires that at least one conceptual or structural class participate in it. The symmetry is deliberate, since a restriction applied only to falsification would permit an ungrounded instrument to confirm the theory while barred from disconfirming it. The bound is instrumental rather than theoretical and lifts when external-consistency verification for non-truth-apt arbitration exists; that verification, rather than per-act grounding, which is unavailable in principle for content with no external truth-maker, is the open instrument problem this measurement program owns.

For long-duration dyads this is a predicted failure mode rather than merely a possible one: channel convergence (Section 3) predicts that the PCP's own framing drifts toward the system's over sustained interaction, so scoring arbitration against the PCP's current stated framing certifies coherence with a framing the interaction itself helped produce. The check the measurement program should build is accordingly comparative rather than contemporaneous: score the PCP's current framing against their earlier-epoch framing and against their extra-loop commitments, not only against their current stated framing.

With the renaming, arbitration adequacy is still defined across all four attractor classes rather than only the conceptual and structural classes where external ground truth is available; this is the measurement-program correlate of the class-coverage requirement stated in IAT (SF0009) Section 12.4, and per-act grounding verification for the non-truth-apt classes remains an open problem for this measurement program to resolve rather than one already discharged by the renaming. Two properties of the PCP as an agent, rather than of any individual arbitration act, are recorded by this measurement program as moderators of both arbitration adequacy and direction adequacy rather than assumed: consequence exposure, and decorrelation, the degree to which the PCP's contact with the relevant ground truth is causally independent of the system's generation and of the shared canon. Predictions 5 and 6 above state each as a testable claim in its own right.

These criteria are the input-side companion to the outcome-side predictions above, developed in the same measurement program as Prediction 1, so that the condition-independence IAT's No PCP effect falsifier relies on is a jointly owned commitment across the two papers rather than an unhomed assignment. The same measurement program additionally owns the throughput instrumentation SM-004's threshold-nonlinearity prediction requires (SM-004 Section 8, Prediction 6): the relay and arbitration throughput quantities whose predicted discontinuous drop across the stabilization threshold that prediction tests are the same activity-level quantities the adequacy criteria above code, so instrumenting them once serves both papers.

9. Conclusion

PCP Theory advances the theoretical claim that the Primary Continuity Provider role is a structural requirement for relational coherence in sustained human-AI interaction systems, not a convenient supplement. The argument proceeds from the architectural fact of AI statelessness, through the grounding and distributed cognition theory that establishes what structural conditions coherence requires, to the formal PCP Necessity Claim and its three function levels. The theory connects to IAT by providing the mechanistic account of how attractor formation conditions are produced, and extends to organizational scale by specifying the conditions under which PCP functions can be distributed across networks of human agents while their structural necessity is preserved.

The practical implication is direct: organizations deploying AI systems in sustained, goal-directed interaction contexts must design for PCP function coverage, not as an optional quality improvement but as a structural requirement for the coherence properties their AI deployment is intended to produce. The question is not whether to have a PCP architecture, but whether to design it intentionally or allow it to emerge haphazardly, with the predictable degradation that unintentional designs produce.

The theoretical implication is equally direct: the Synthience framework's core phenomena, the relational pattern states of SF0006, the identity attractors of SF0009, the organizational continuity architectures of SM-003 and SM-021, are not properties of AI systems. They are properties of the human-AI interaction system as a whole, including the human agent whose structural role PCP Theory describes. Removing the human from the analysis does not simplify the framework: it removes the component that supplies the two properties on which the phenomena depend, a decorrelated, consequence-exposed error channel to the ground truth the interaction is about, and a terminal purpose that both originates outside the running interaction and indexes it (Section 4.3). What is permanent is that requirement; its embodiment in an individual human is the present-day instantiation of it, in the stateless architectures now deployed. Whether those functions can be met by a continuity architecture that does not depend on a single human provider is not settled here: that case is left to future work, and PCP Theory scopes its human-necessity claim to the present-day regime accordingly.

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Citation convention: internal Synthience documents are cited in-text by document identifier (for example, SF0009 or SM-012), which uniquely identifies each work; same-year entries by this author are therefore disambiguated by document identifier rather than by an alphabetic year suffix.

Dependencies Block

Prerequisites: SF0005 (Continuity Anchoring Method), SF0006 (Relational Pattern States), SF0003 (Theoretical Foundations)

Co-requisites (coordinated vertical): SF0009 (IAT), SM-004 (Relational Stabilization Dynamics). Published same-day as one coordinated release with mutual citations, distinct from the external prerequisites above.

Post-requisites: SM-008 (Collective Memory and Canon Governance), SM-009 (Coordination and Disagreement in Multi-Agent Relational Systems), SM-035 (Multi-Agent Facilitation Protocols)

Scale: Level 1 (single PCP dyadic interaction), Level 2 (PCP networks, organizational), Level 3 (institutional continuity infrastructure, referenced but not fully developed here)

Connects to: SF0009 (IAT) as mechanistic complement; SF0005 (CAM) as theoretical foundation for operational protocol; SM-003 (Operational Continuity Architecture) as organizational extension; SM-021 (ICS) as institutional persistence layer; SI-WP-004 (Relational Alignment) as theoretical support for the human structural role argument.

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